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METRIC

DOD-STD-2133 (SH)
3 AUGUST 1981

DEPARTMENT OF DEFENSE
DESIGN CRITERIA

CABLE ARRANGEMENT FOR MINIMUM SPRAY MAGNETIC FIELD (METRIC)

AMSC N/A FSC 6145

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DEPARTMENT OF THE NAVY

NAVAL SEA SYSTEMS COMMAND

WASHINGTON, DC 20362

Cable Arrangement for Minimum Stray Magnetic Field

DOD-STD-2133(SH)

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1. SCOPE

1.1 Scope. This standard covers the arrangement of power cables in a manner that will reduce to a minimum the stray magnetic field produced by the current in these cables.

1.2 Application. Requirements specified herein are applicable to shipboard installation of direct current (d.c.) and alternating current (a.c.) power cables aboard minesweepers and other special installations where stray magnetic fields must be reduced to a minimum.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following document, of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
803-5001027 - Electric Plant Installation Standard Methods.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 Crossways cable run terminal approach. A cable run approaches its connecting terminals in a crossways direction when it approaches at a vertical angle of 90 degrees from the plane common to all terminals (see figure 1).

3.2 Current loop. A current loop is a closed electric conductor. It may have one or more turns of any size or shape conductor and may be arranged in any way. A simple current loop is a closed conductor making one turn in a single plane, or alternatively, making a number of turns which are in the same plane or in parallel planes and so close together that, to a first approximation, they can be considered to be physically coincident in space. More complicated current loops can be resolved into a combination of simple current loops.

3.3 Edgewise terminals. Edgewise terminals are oriented such that their widths lie in parallel vertical planes (see figure 1).

3.4 Endways cable run terminal approach. A cable run approaches its connecting terminals in an endways direction when it approaches the forward edges of the terminals at an angle of zero degrees (see figure 1).

3.5 Flat terminals. Flat terminals are oriented such that their widths lie in the same horizontal plane (see figure 1).

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3.6 Flux density. Flux density is that magnetic flux per unit area normal to the direction of flux. It is also known as magnetic induction. The usual symbol for flux density is B. The unit of flux density is tesla (T). One tesla equals one weber per square meter (Wb/m^2).

3.7 Magnetic minesweep cable. The magnetic minesweep cable is the interconnecting cable between the magnetic minesweep generator(s) and the minesweep cable terminal box.

3.8 Magnetic minesweep terminal box. The magnetic minesweep terminal box is the device which enables the magnetic minesweep cable to be connected to the sweep tail cable.

3.9 Sideways cable run terminal approach. A cable run approaches its connecting terminals in a sideways direction when it approaches the forward edges of the terminals at a horizontal angle of 90 degrees (see figure 1).

3.10 Stray magnetic field. A stray magnetic field is a magnetic field produced by an electric circuit or equipment which is undesirable and external to the electric circuit or equipment causing it.

3.11 Sweep tail cable. The sweep tail cable is the cable aft of the magnetic minesweep terminal box. When this cable is not in use, it is stored on the magnetic minesweep cable reel.

4. GENERAL REQUIREMENTS

4.1 Direct current (d.c.) power cables.

4.1.1 Four-conductor, quadded cables. Four-conductor, quadded cables shall be utilized for d.c. power cables, wherever practicable.

4.1.2 Utilization of double-conductor cables. Where four-conductor, quadded cable is unavailable or not practicable, one double-conductor cable may be utilized in a d.c. cable run if it is twisted such that the cable's conductors are transposed at regular intervals by the twist of the conductors to form the lay of the cable or the conductors are concentric and the size of the cable is small enough to handle conveniently for the selected application. Two or more double-conductor cables connected in parallel shall not be utilized in d.c. cable runs since unequal division of current between conductors connected in parallel can give rise to a magnetic field of objectionable magnitude.

4.1.3 Utilization of single-conductor cables. An even multiple of parallel, single-conductor cables in the positive and negative branches of a d.c. cable run shall be utilized. The cables shall be arranged in accordance with section 5. Cable runs consisting of only one or an odd multiple of single-conductor cables in the positive and negative branches of a d.c. circuit shall not be utilized since an odd number of current loops makes magnetic field compensation by arrangement impracticable and results in large magnetic fields (see figure 2).

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4.2 Alternating current (a.c.) power cables.

4.2.1 Phase conductors in a common cable. Special cable arrangement is not necessary when all phase conductors of an a.c. circuit are in the same cable.

4.2.2 Phase conductors in separate cables. Where it is necessary to install a.c. phase conductors in separate cables, these cables shall not be grouped on the same cable hangers with cables carrying d.c. and shall be as close together as practicable throughout the entire length of the cable run. Closed loops of magnetic material around a phase conductor cable and the placement of any magnetic material between the phase conductor cables shall be avoided.

5. DETAILED REQUIREMENTS

5.1 Cable arrangements.

5.1.1 Arrangement of cables for opposing current loops. Parallel, single-conductor cables in the positive and negative branches of a d.c. cable run shall be arranged in a manner that will result in opposing (nonadditive) current loops, so that their magnetic fields are opposed and self-compensating. Figure 2 illustrates typical cable arrangements which are preferred and nonpreferred and their relative resulting magnetic flux density values.

5.1.2 Equal current division among cables connected in parallel. Parallel, single-conductor cables in the positive and negative branches of a d.c. cable run shall be selected and installed in a manner to ensure that the resistance of each branch circuit in parallel is as nearly equal as possible so that circuit current divides equally among all of the conductors connected in parallel. Figure 3 illustrates the effects of unequal current division in cables connected in parallel.

5.1.3 Spiraling the cable run to minimize the effects of unequal current division. A d.c. circuit cable run array consisting of multiple, single-conductor cables shall be spiraled (twisted) on itself from one cable hanger to the next so that any magnetic field created by an unequal current division in parallel conductors of the circuit is compensated for. Cable runs less than 1.5 meters long need not be spiraled; spiraling will aid in minimizing the separation between adjacent conductors of the cable. Figure 4 illustrates a typical spiraling arrangement for a d.c. circuit cable array composed of four cables in a quad array and eight cables in a circular array.

5.1.4 Cable spacing. Multiple, single-conductor d.c. cables in a cable run shall be spaced equally to each other along their entire run so that the opposing current loops created by the cable arrangements will result in a net magnetic moment as close to zero as possible. Cables in a quad arrangement shall be pressed tightly together all along the cable run with their centers at the corners of a square. Figure 5 illustrates the

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typical effects of excessive cable separation in a quad arrangement. Equal spacing of the cables in a circular array arranged about a central core shall be accomplished by utilizing cables of the same diameter, selecting a central core of the proper diameter, and by tightly clamping the cables against the core.

5.2 Magnetic minesweep cable arrangements.

5.2.1 Four-cable arrangements. Magnetic minesweeping cable arrangements consisting of two parallel, single-conductor cables in both the positive and negative branches of the cable run, for a total of four cables, shall be arranged to form a quad array. The cable centers of the quad shall form a square with alternate positive and negative polarity (see figure 2A). The arrangement and installation of the cables shall be in accordance with 5.1 with the additional requirement that the spiraling of the quad shall be a right hand lay of 360 degrees every 1.0 to 3.0 meters (see figure 4A). With the exception of terminal connections and hull penetrations, the separation, if any, of the outer sheath of adjacent conductors of the quad should be less than 12.5 mm.

5.2.1.1 Bulkhead penetrations. Where space permits, a single community type stuffing tube shall be used for the cables of a quad when penetrating a watertight bulkhead. Details of the arrangement shall be in accordance with the standard method for community stuffing tubes penetrating watertight wood bulkheads on minesweepers as shown on NAVSEA Drawing 803-5001027. Where the cables of the quad penetrate a bulkhead through individual stuffing tubes, the separation of the outer sheath of adjacent cables shall not exceed 75 mm at the bulkhead and 12.5 mm at a distance of 450 mm from the bulkhead. The cables shall be tightly bound together just prior to entering and after leaving the stuffing tubes.

5.2.2 Eight-cable arrangements. Magnetic minesweeping cable arrangements consisting of four parallel, single-conductor cables in both the positive and negative branches of the cable run, for a total of eight cables, shall be arranged around a central core as shown on figure 6. The arrangement, installation and spiraling of the cables shall be in accordance with 5.1.

5.2.3 Cable arrangements with other than four or eight cables. When the magnetic minesweep cable installation design is such that an even multiple of parallel, single-conductor cables other than two (four total) or four (eight total) is required in both the positive and negative branches of the cable run, the cables shall be arranged about a central core with the cables alternately positive and negative. For convenient connection to three-terminal equipment, the preferred total number of cables is an integral multiple of four.

5.2.4 Sweep tail cable. In order to prevent the formation of excessive current loops, when connecting the sweep tail cable to the magnetic minesweep terminal box, the sweep tail conductors shall have a separation of not more than 12.5 mm at a distance of 300 mm from the terminal connection. After the cable is connected to the terminal box, the conductors shall be tightly bound in order to prevent further separation between conductors of the cable during the sweep operation.

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5.3 Cable terminal connections.

5.3.1 Compatibility of cable runs and terminal. The terminals on equipment shall be arranged for ready connection to the cable runs which take current to and from the equipment. In the absence of a terminal arrangement to which the cable run can be conveniently connected, an adapter between the cable run and the equipment shall be utilized.

5.3.2 Arrangement of terminals and approach by cable run. Terminals for cable runs shall be equally spaced, parallel terminals in the same plane arranged edgewise or flat, with the cable run approaching the terminals endways, crossways, or sideways. Edgewise terminals are preferred since the center-to-center distance between terminals can be made smaller than for flat terminals, which results in smaller current loops. Edgewise terminals can utilize the same type of cable lugs for endways and crossways connections.

5.3.3 Three-terminal arrangements. Three-terminal arrangements are preferred and shall be in accordance with figure 1. Cable conections to the terminals shall be made in accordance with 5.3.3.1 through 5.3.3.7. Endways, crossways, and sideways orientation of the cable run approach is illustrated on figure 1. If figure 1 is rotated to bring the terminals into some other plane, or in some other orientation in the same plane, the endways, crossways, and sideways directions rotate with the figure.

5.3.3.1 Four-cable connection with endways approach to edgewise terminals. A four, single-conductor cable run approaching endways to edgewise terminals shall be connected in accordance with figure 7.

5.3.3.2 Four-cable connection with crossways approach to edgewise terminals. A four, single-conductor cable run approaching crossways to edgewise terminals shall be connected in accordance with figure 8.

5.3.3.3 Four-cable connection with endways approach to flat terminals. A four, single-conductor cable run approaching endways to flat terminals shall be connected in accordance with figure 9.

5.3.3.4 Four-cable connection with crossways approach to flat terminals. A four, single-conductor cable run approaching crossways to flat terminals shall be connected in accordance with figure 10. With flat terminals, different types of cable lugs will have to be used for crossways connections rather than those used with edgewise connections.

5.3.3.5 Four-cable connection with sideways approach to edgewise or flat terminals. A four, single-conductor cable run approaching sideways to edgewise or flat terminals shall be avoided since special bus bar adapters are required to make a connection that will result in a minimum stray magnetic field.

5.3.3.6 Eight-cable connection. An eight, single-conductor cable run approaching endways, crossways, or sideways to flat terminals shall be connected to the terminals in accordance with figures 11, 12, and 13, respectively. In each case, the positive cables shall be located

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symmetrically about the axis of the cable run; this shall also be true of the negative cables. The general scheme shown shall also be applicable to edgewise terminals.

5.3.3.7 General connection method for six cables. An endways or crossways connection to three terminals for a cable run consisting of six, single-conductor cables about a central core may be made as follows (see figure 14):

(a) Draw in every other cable and connect it to a single conductor which is on an extension of the cable axis. This single conductor shall than be connected to the central terminal of the equipment.
(b) Connect the remaining three cables to a conducting ring which is concentric with the cable axis.
(c) Connect two diametrically opposite points on this ring to the two outside terminals of the equipment.

This arrangement will create current loops of equal area and opposite polarity, causing the stray magnetic field to be small. A simple modification of the above procedure can accommodate a sideways cable run.

5.4 Minesweep cable terminal connections.

5.4.1 Terminal and disconnect switch connections for minesweep generator cable runs with four single-conductor cables. Terminal and disconnect switch connections for minesweep generator cable runs with four single-conductor cables shall be arranged in accordance with figures 15, 16 or 17. The cables shall be tightly bound together as close as possible to the terminal lugs and the terminal box connection. The separation of the outer sheath of adjacent cables shall not exceed 12.5 mm at a distance of 450 mm from the bulkhead.

5.4.2 Terminal and disconnect switch connections for minesweep generator cable runs with eight, single-conductor cables. Terminal and disconnect switch connectins for minesweep generator cable runs with eight, single-conductor cables shall be arranged in accordance with figure 18.

5.5 Minesweep cable run and terminal inspection. A visual inspection of cable runs and connecting terminal installation arrangements shall be made to ensure that the cable is arranged in accordance with the requirements of this standard. The inspection shall start at the minesweep generator and shall include all minesweep cables to the sweep tail terminal box connection. Particular attention should be given to the terminal connections, bulkhead piercing and bends in the cable. A check shall be made at each terminal connection to ensure that the polarity of the cable is arranged so that all adjacent conductors have opposite polarities. In particular, the following inspections shall be made:

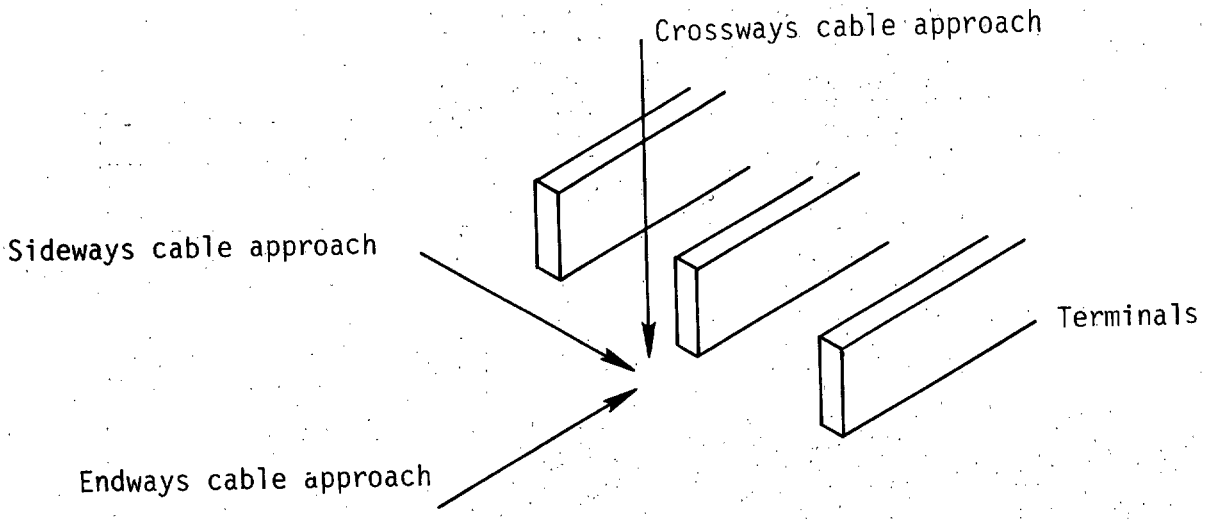
(a) Inspect minesweep generator(s) terminal connections for:
    (1) Proper arrangement of lugs.
    (2) Cable tightly bound near the terminals.
    (3) Cable conductors arranged in accordance with this standard.

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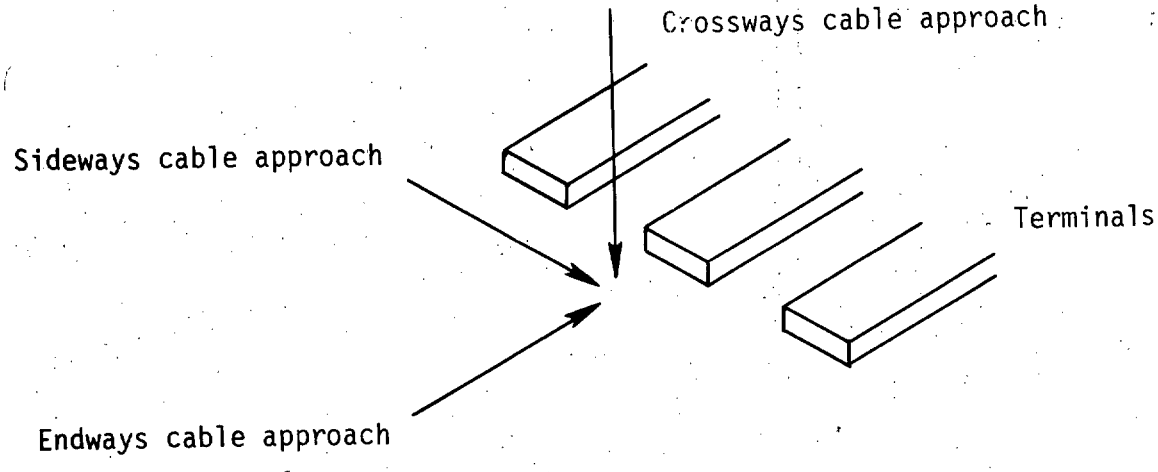
(b) Inspect the cable from generator(s) to disconnect switch for:
    (1) Excessive separation in the cable conductors.
    (2) Adequate twisting of cable.
(c) Inspect disconnect switch terminal connections for:
    (1) Proper arrangement of lugs.
    (2) Cable tightly bound near terminals.
    (3) Cable conductors arranged in accordance with this standard.
(d) Inspect the cable penetration of watertight bulkhead for:
    (1) Minimum spacing of stuffing tubes.
    (2) Cable tightly bound just prior to entering, and after leaving the stuffing tubes.
(e) Inspect cable from disconnect switch to terminal box for:
    (1) Excessive separation in the cable conductors.
    (2) Adequate twisting of cable.
(f) Inspect sweep tail cable to terminal box for:
    (1) Excessive separation of the sweep tail cable.
    (2) Adequate twisting of cable.
    (3) Cable tightly bound together.
    (4) Proper positive-negative connection of conductors of sweep tail cable to terminal box connectors.

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A. EDGEWISE ARRANGEMENT



B. FLAT ARRANGEMENT

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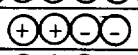
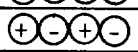
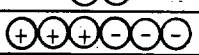
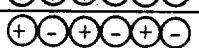
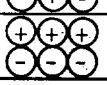
FIGURE 1. Edgewise and flat three-terminal arrangements.

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A. Typical preferred cable arrangements.

Cable arrangement	Max. magnetic flux density-nanotesla
	2.0
	1.0

B. Typical non-preferred cable arrangements.

Cable arrangement	Max. magnetic flux density-nanotesla
	90
	570
	280
	280
	150
	850
	280
	50
	150

SH 11942

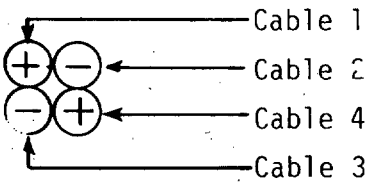
NOTES:

1. Cable arrangement figures illustrate the cross-sectional views of the single conductor cables in a cable run. "+" indicates positive conductor cables connected in parallel. "-" indicates negative conductor cables connected in parallel. For a given arrangement, if all "+" are changed to "-" and all "-" changed to "+", the same indicated maximum flux density will result.
2. The indicated maximum flux density is the vertical component calculated on a plane 7.5 meters above or below the cable run based upon cables with an outside diameter of 40 mm that are touching each other so that the center-to-center distance between adjacent cables is 40 mm. The length of the cable run is 18 meters and the direct current magnitude is 2,000 amperes (A) (1,000 A per cable for cable runs with two cables in each branch and 667 A per cable for cable runs with three cables in each branch). No magnetic materials are assumed to be present anywhere in the region around the cable run and the total current is assumed to be equally divided between conductors connected in parallel.

FIGURE 2. Arrangement of single conductor d.c. cables for opposing current loops.

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Cable arrangement:



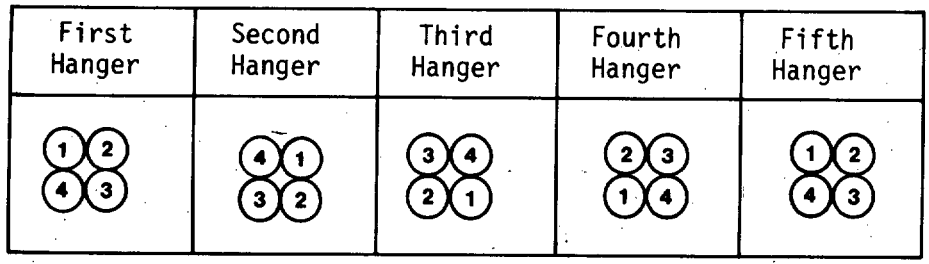
Current division among conductors (amperes)				Maximum magnetic flux density (nanotesla)
Cable 1	Cable 2	Cable 3	Cable 4	
1,000	1,000	1,000	1,000	1
2,000	2,000	0	0	280
0	1,000	1,000	2,000	150
900	900	1,100	1,100	30

Refer to notes of Figure 2.

SH 11943

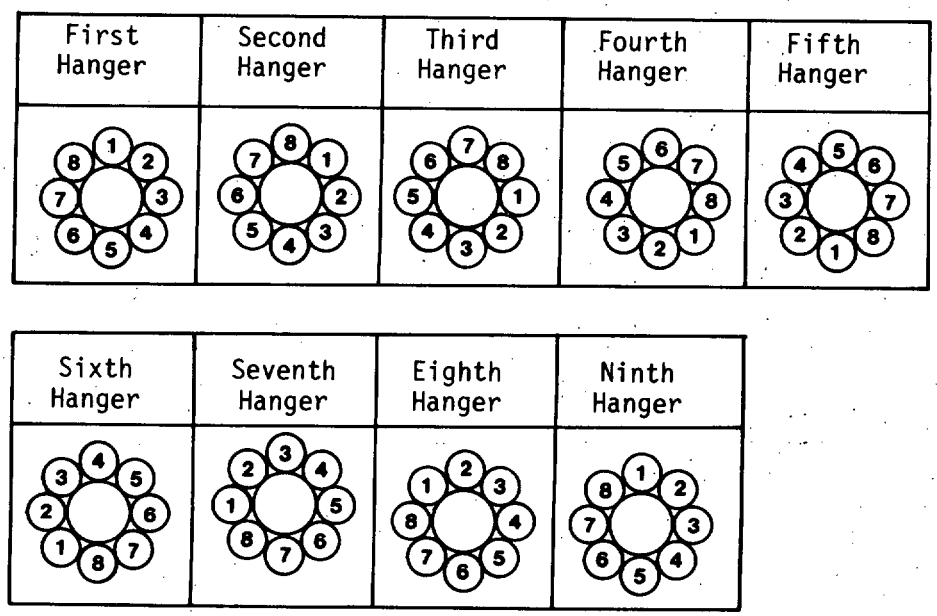
FIGURE 3. Effects of unequal current division in single-conductor cables connected in parallel.

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A. Four cable array configuration at five consecutive cable hangers (cross-sectional views)

NOTE:
1, 2, 3, and 4 denote each of the four cables in a quad array. Each cable shifts 90 degrees from one cable hanger to the next.

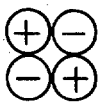
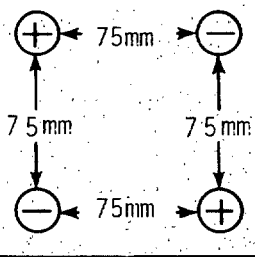
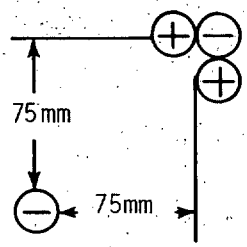


B. Eight cable circular array configuration at nine consecutive cable hangers (cross-sectional views)

SH 11944

FIGURE 4. Typical spiraling arrangements for d.c. circuit cable array.

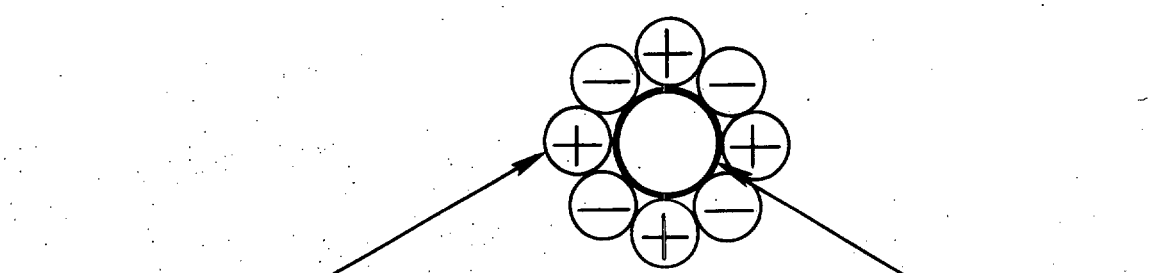
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Cable arrangement	Maximum magnetic flux density (nanotesla)
	0.42
	2.62
	31.4

NOTE:
The indicated maximum flux density is calculated at a distance of 7.5 meters based upon cables with an outside diameter of 50 mm. The length of the cable run is 1.25 meters and the direct current magnitude is 1,500 A (750 A per cable).

FIGURE 5. Typical effects of excessive cable separation in a quad arrangement.

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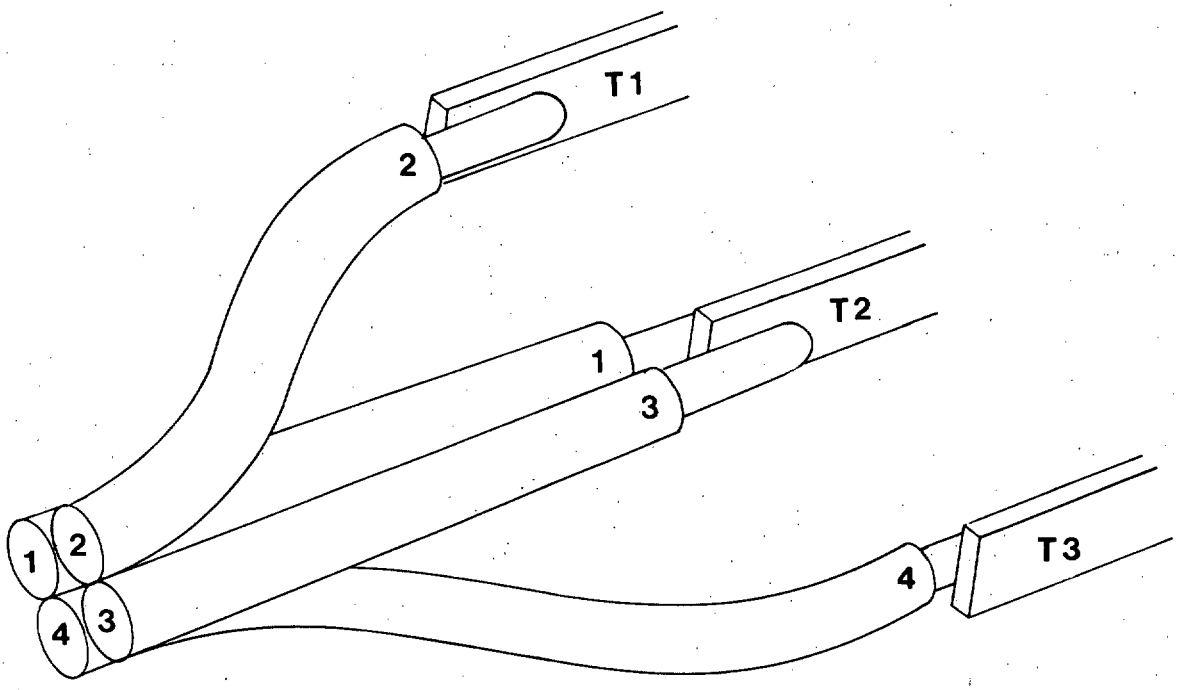


CROSS-SECTIONAL VIEW

SH 11946

FIGURE 6. Magnetic minesweep eight cable arrangement.

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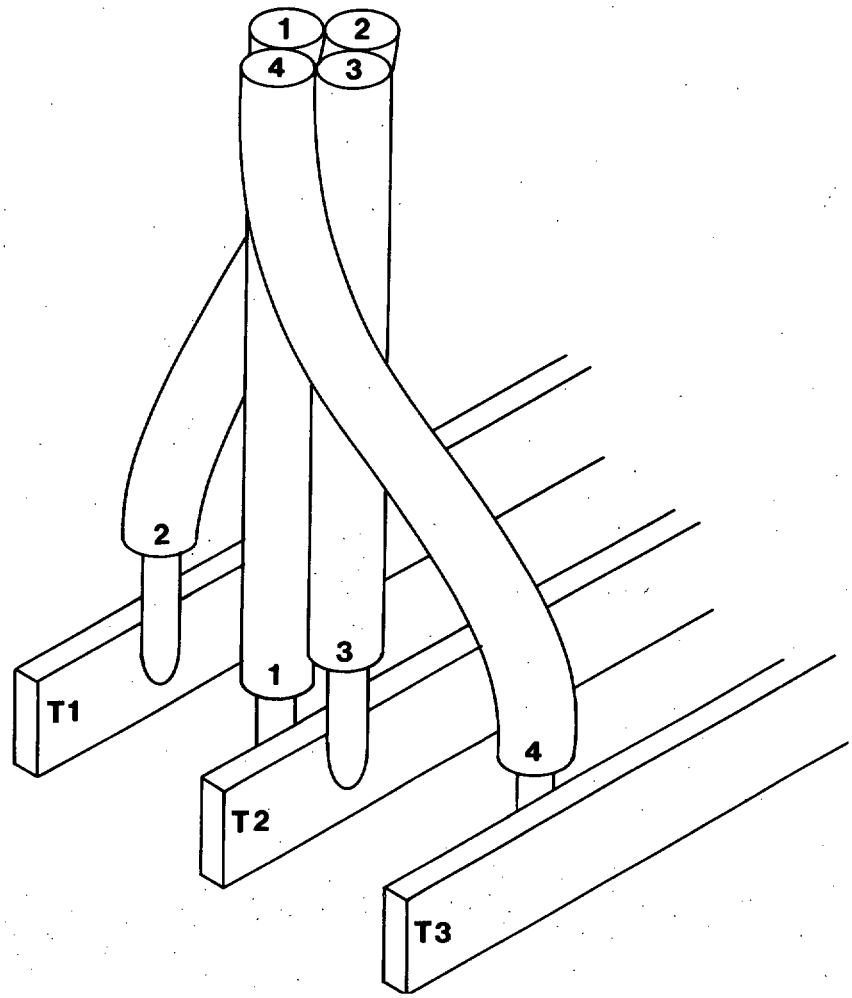
SH 11947

NOTES:

1. The cable run shall approach the terminals with its axis in line with the horizontal axis of the center terminal, T2.
2. The two cables 1 and 3, which lie in a horizontal plane, shall be bolted to the opposite sides of terminal T2.
3. Cable 2 is bent to the left and slightly down to connect to terminal T1. Cable 4 is bent to the right and slightly up to connect to terminal T3.
4. The area outlined by cables 1 and 4 equals the area of the cable run outlined by cables 2 and 3.
5. The polarity of the current in the loop made by cables 1 and 4 is opposite to the polarity of the current in the loop made by cables 2 and 3.

FIGURE 7. Connection of an endways cable run with four cables to edgewise terminals.

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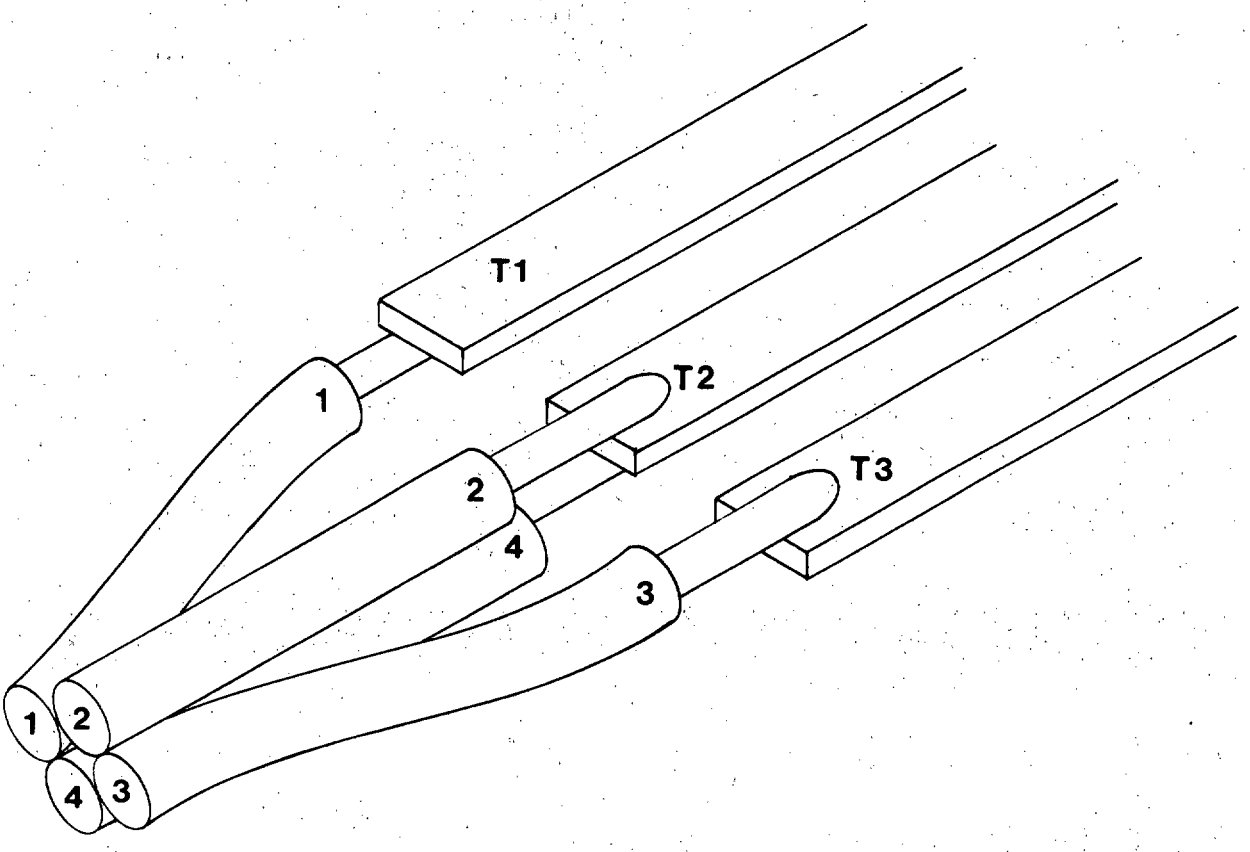
SH 11948

NOTES:

1. This figure can be obtained from figure 7 by rotating through 90 degrees the cable run (or terminals) about an axis through the points of attachment of the cables to the terminals, when these points are on a line perpendicular to the terminals.
2. The cable run shall approach the terminals with its axis in line with the vertical axis of the center terminal, T2.
3. The two cables 1 and 3, which lie in a vertical plane, shall be bolted to the opposite sides of terminal T2.
4. Cable 2 is bent to the left and slightly forward to connect to terminal T1. Cable 4 is bent to the right and slightly back to connect to terminal T3.
5. The area outlined by cables 1 and 4 equals the area outlined by cables 2 and 3.
6. The polarity of the current in the loop made by cables 1 and 4 is opposite to the polarity of the current in the loop made by cables 2 and 3.

FIGURE 8. Connection of a crossways cable run with four cables to edgewise terminals.

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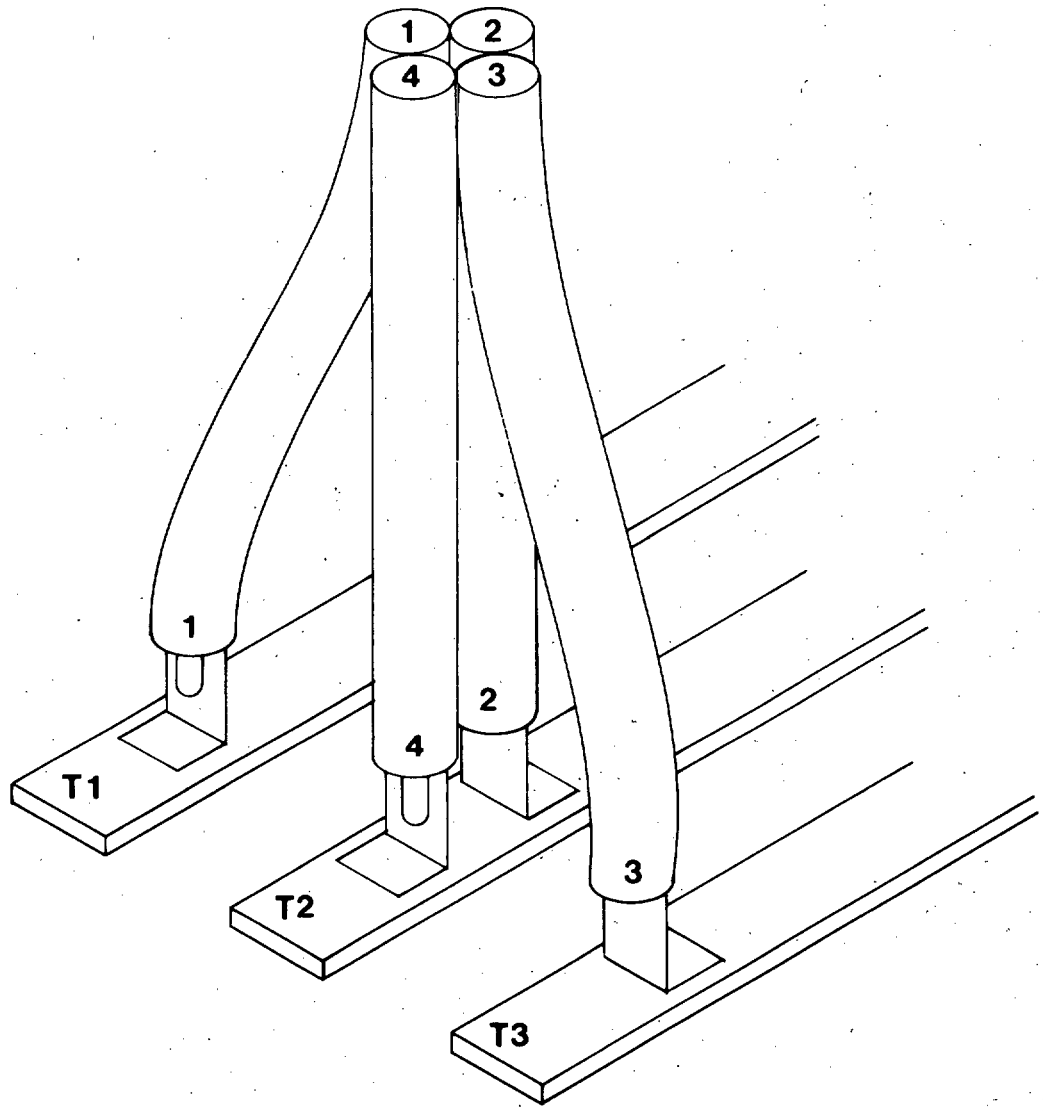
SH 11949

NOTES:

1. The cable run shall approach the terminals with its axis in line with the horizontal axis of the center terminal T2.
2. The two cables 2 and 4 which lie in a vertical plane, shall be bolted to the upper and lower sides of terminal T2.
3. Cable 1 is bent to the left and slightly down to be bolted to the under side of terminal T1. Cable 3 is bent to the right and slightly up to the upper side of terminal T3.
4. The area outlined by cables 1 and 4 equals the area outlined by cables 2 and 3.
5. The polarity of the current in the loop made by cables 1 and 4 is opposite to the polarity of the current in the loop made by cables 2 and 3.

FIGURE 9. Connection of an endways cable run with four cables to flat terminals.

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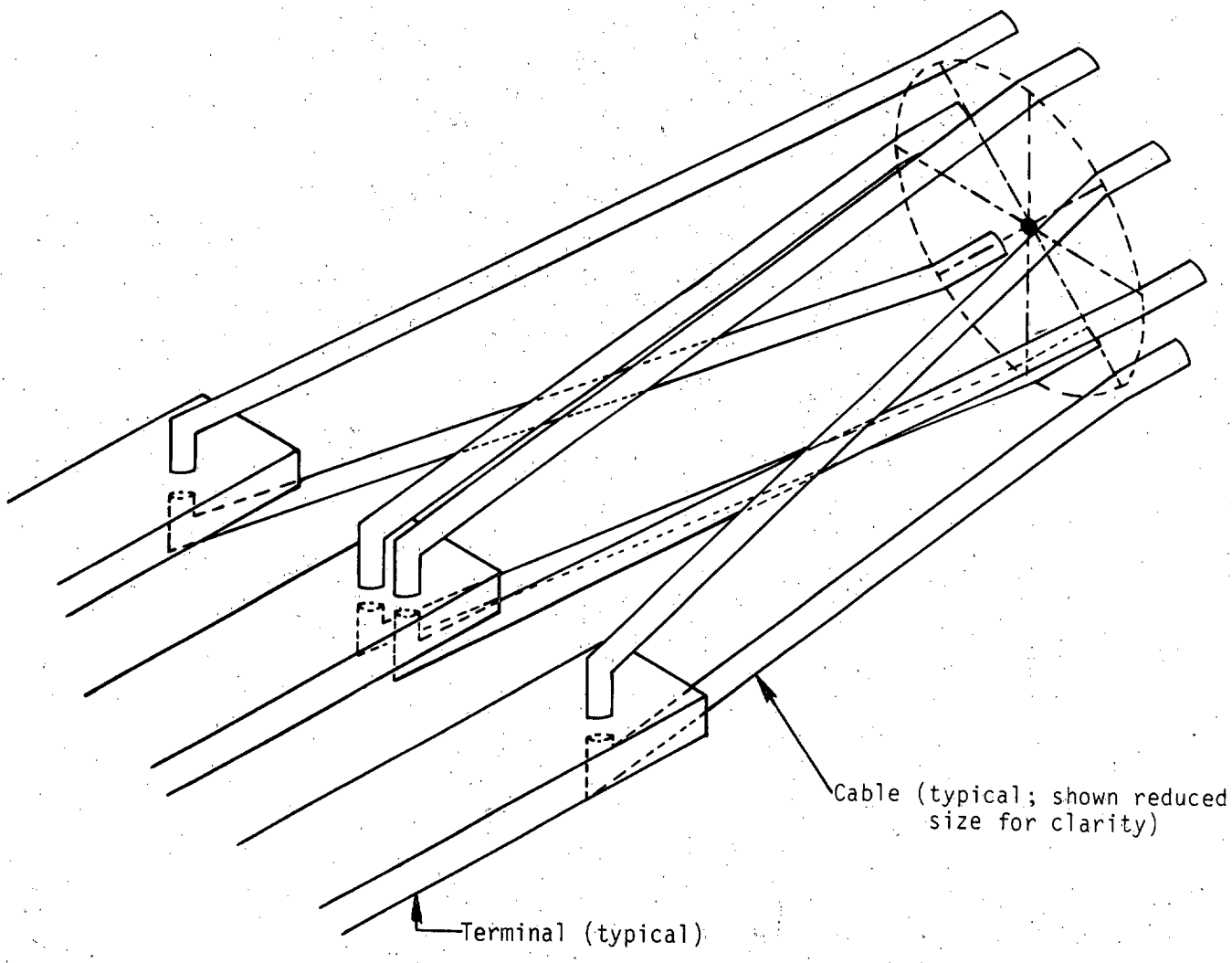
SH 11950

NOTES:

1. The cable run shall approach the terminals with its axis in line with the vertical axis of the center terminal T2.
2. The two cables 2 and 4 which lie in a vertical plane, shall be bolted to the upper side of terminal T2.
3. Cable 1 is bent to the left and slightly forward to be bolted to the upper side of terminal T1. Cable 3 is bent to the right and slightly back to the upper side of terminal T3.
4. The area outlined by cables 1 and 4 equals the area outlined by cables 2 and 3.
5. The polarity of the current in the loop made by cables 1 and 4 is opposite to the polarity of the current in the loop made by cables 2 and 3.

FIGURE 10. Connection to a crossways cable run with four cables to flat terminals.

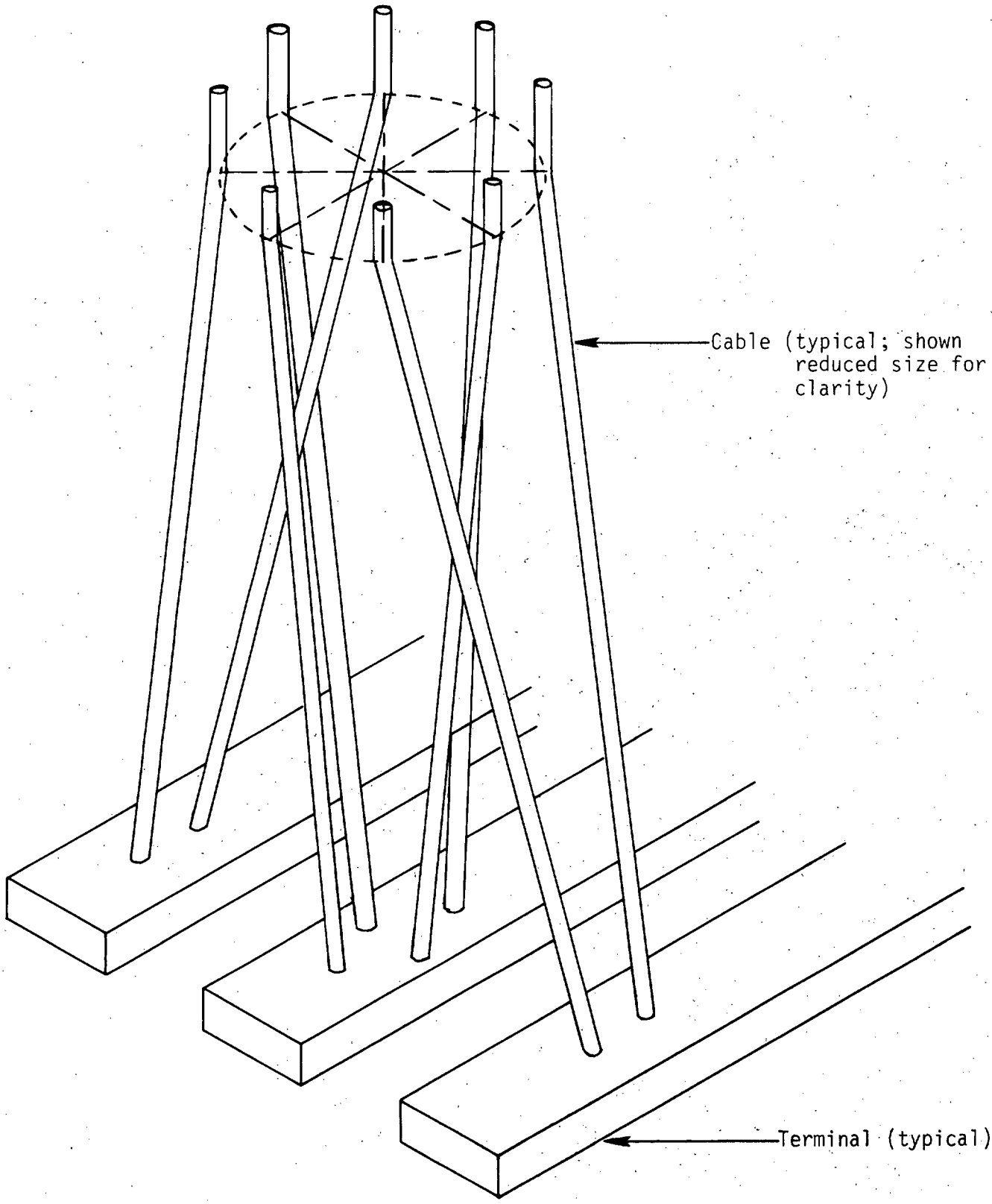
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SH 11951

FIGURE 11. Connection to an endways cable run with eight cables to flat terminals.

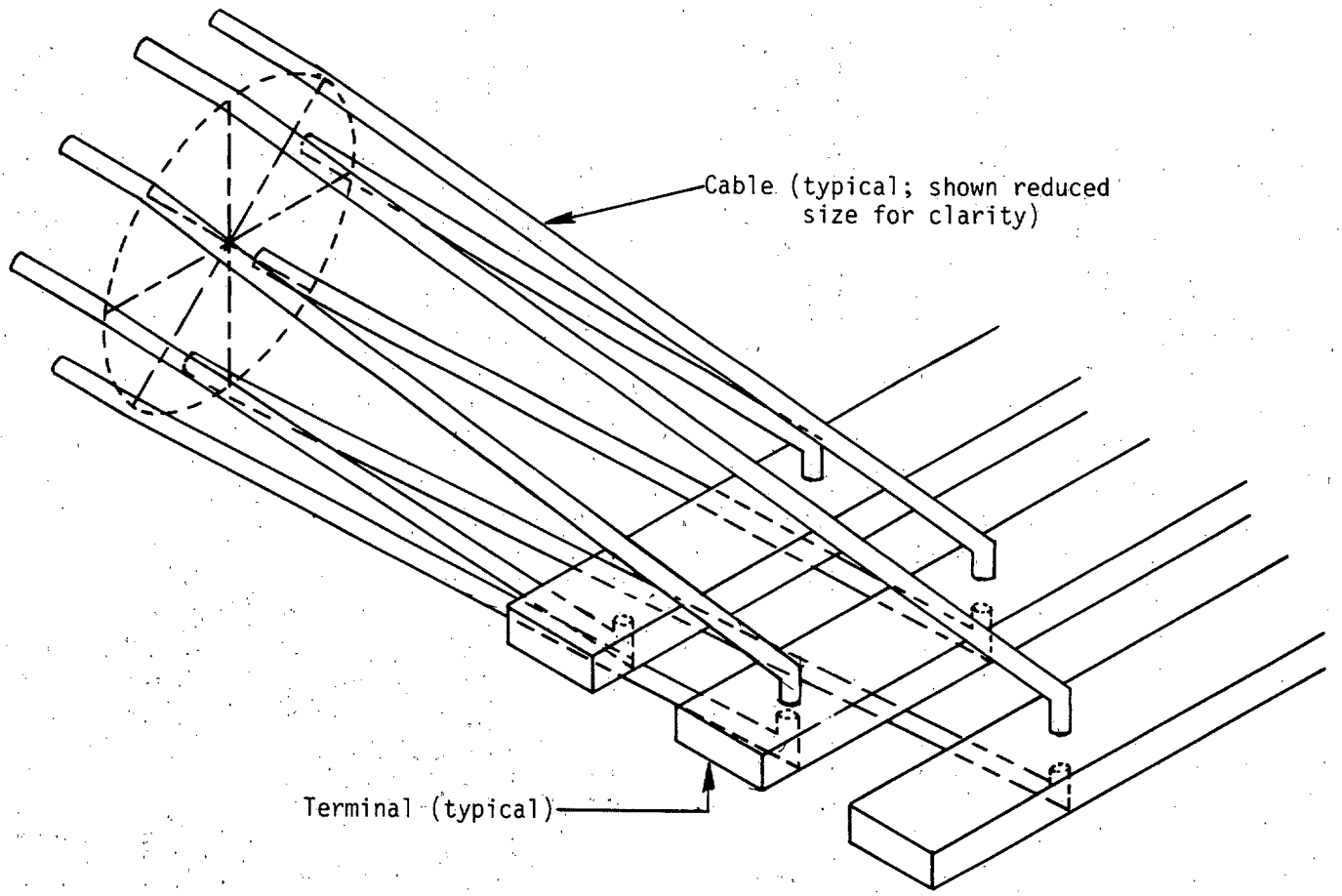
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SH 11952

FIGURE 12. Connection to a crossways cable run with eight cables to flat terminals.

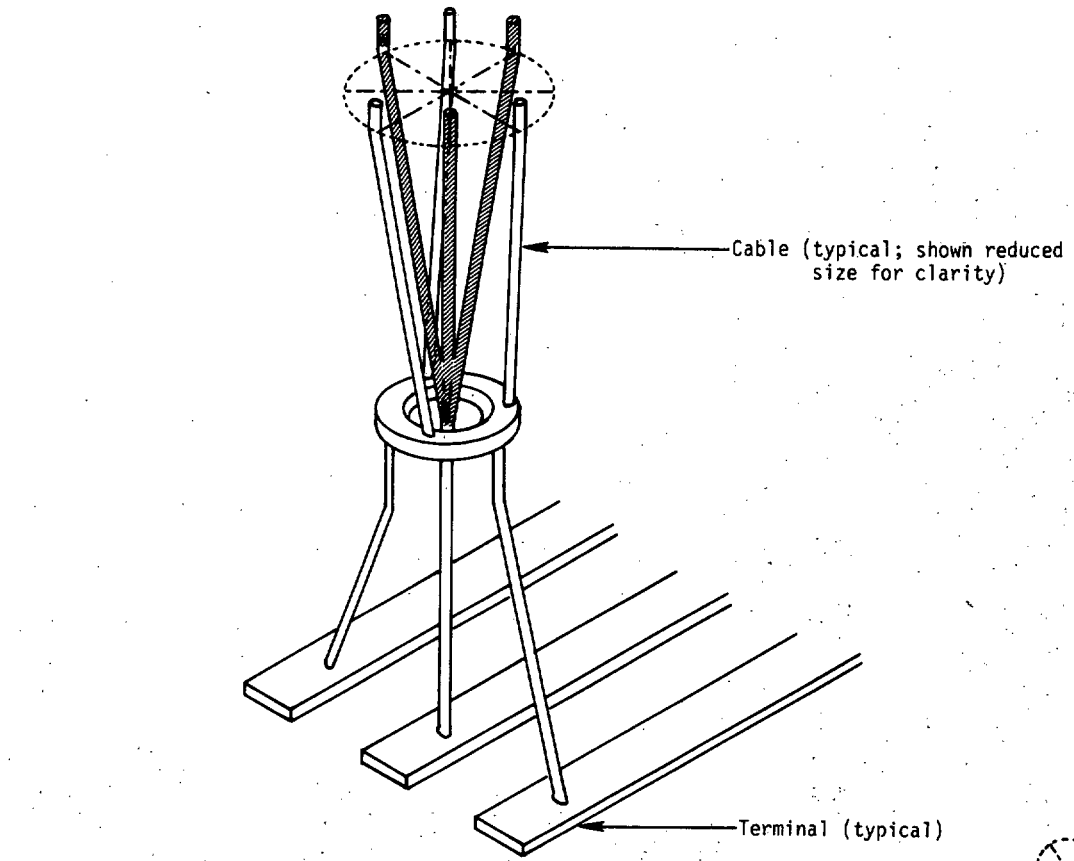
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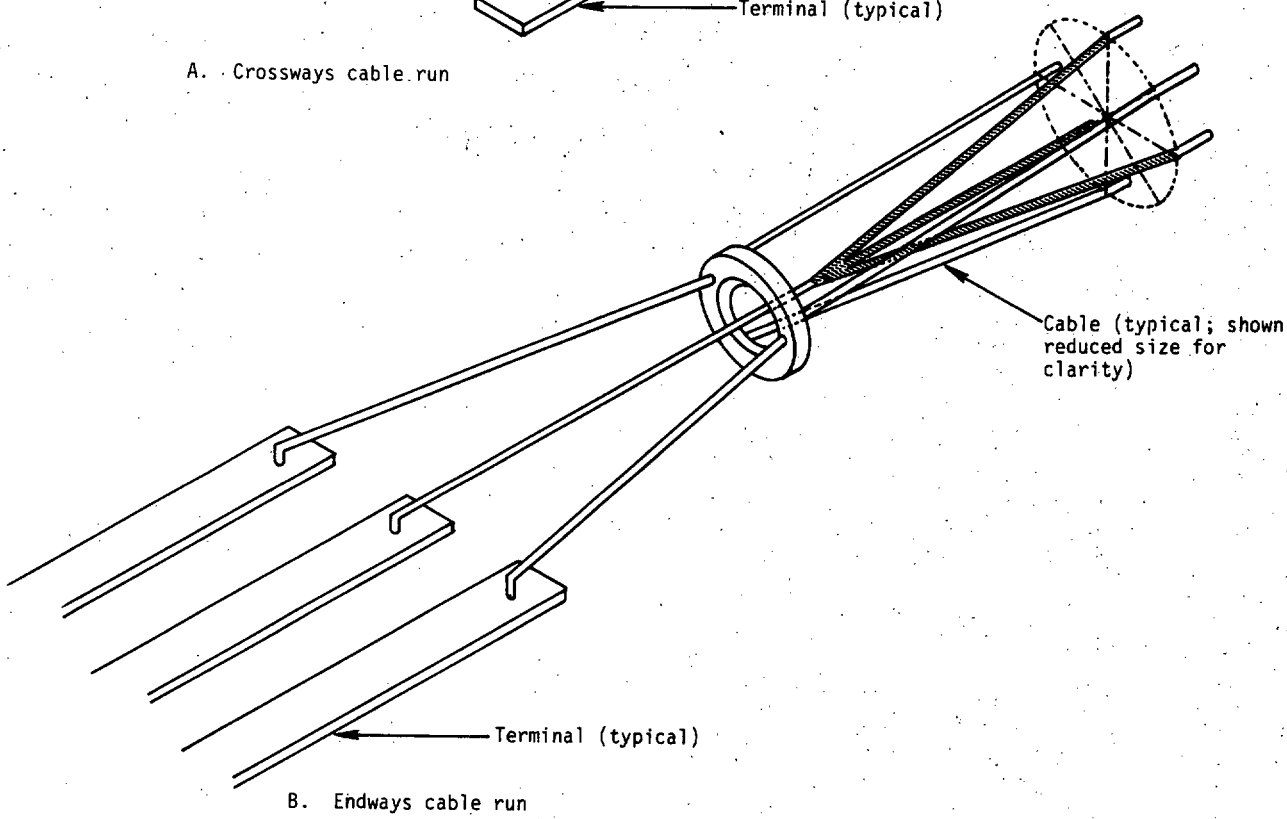
SH 11953

FIGURE 13. Connection to a sideways cable run with eight cables to flat terminals.

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A. Crossways cable run

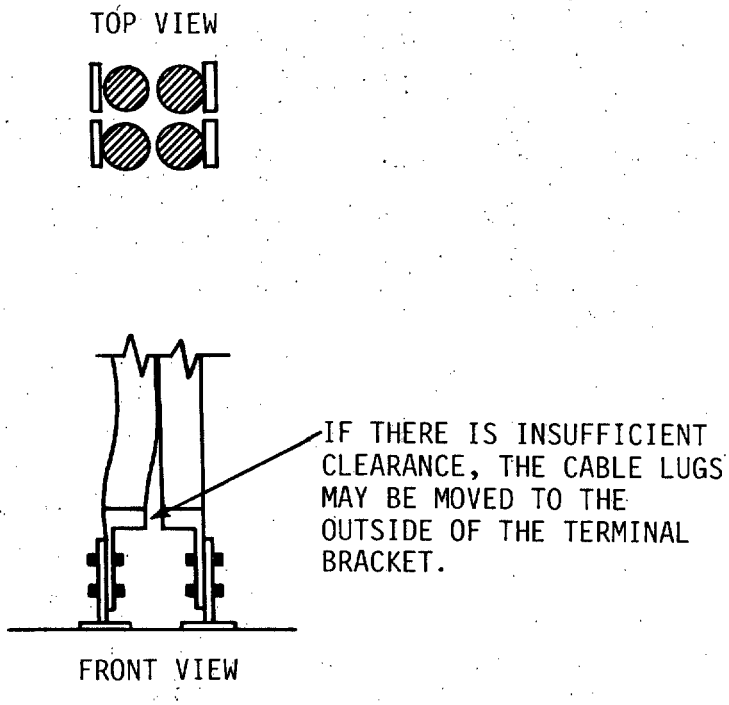


B. Endways cable run

SH 11954

FIGURE 14. Connection to an endways and crossways cable run with six cables.

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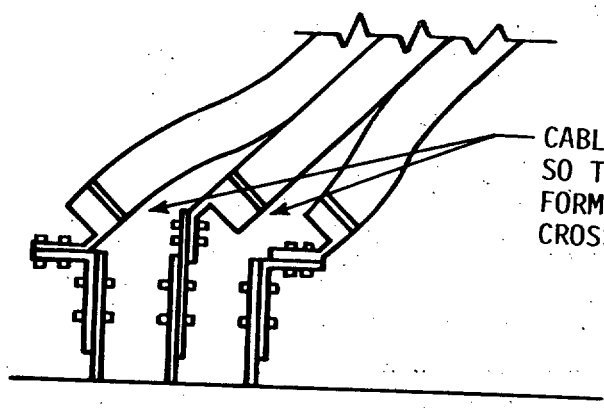
SH 11955

FIGURE 15. Four-cable, four-terminal blade connection.

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TOP VIEW



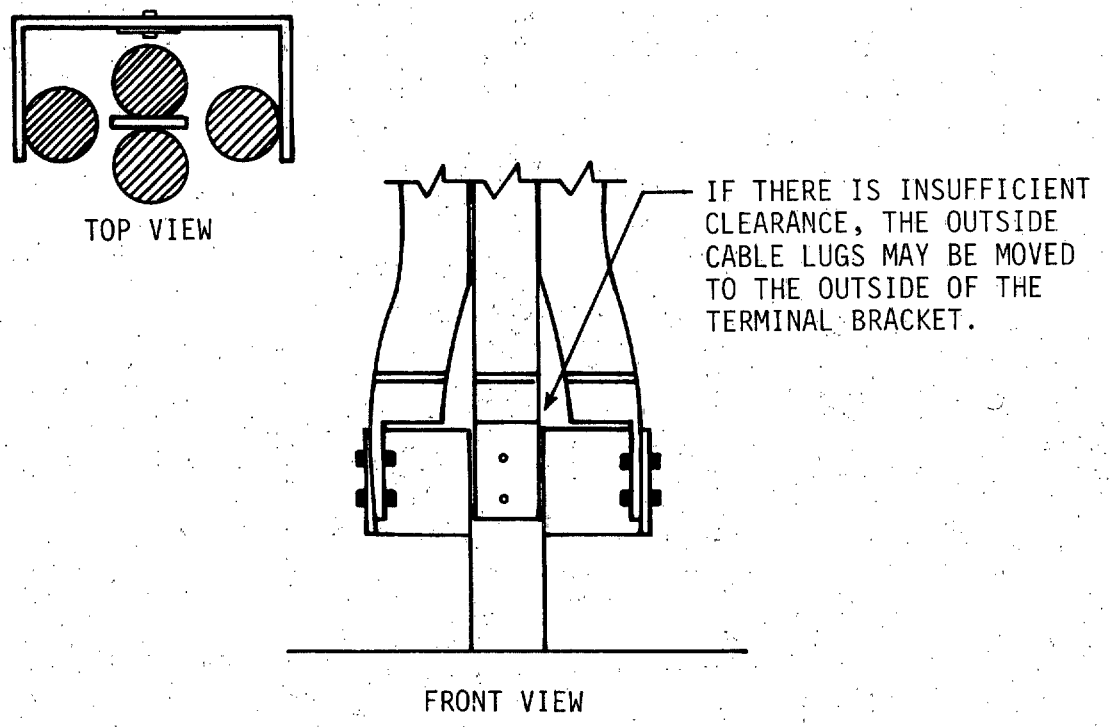
FRONT VIEW

SH 11956

NOTE:
Cables leave at approximately 45 degree angle to blades.

FIGURE 16. Four-cable, three-terminal blade connection.

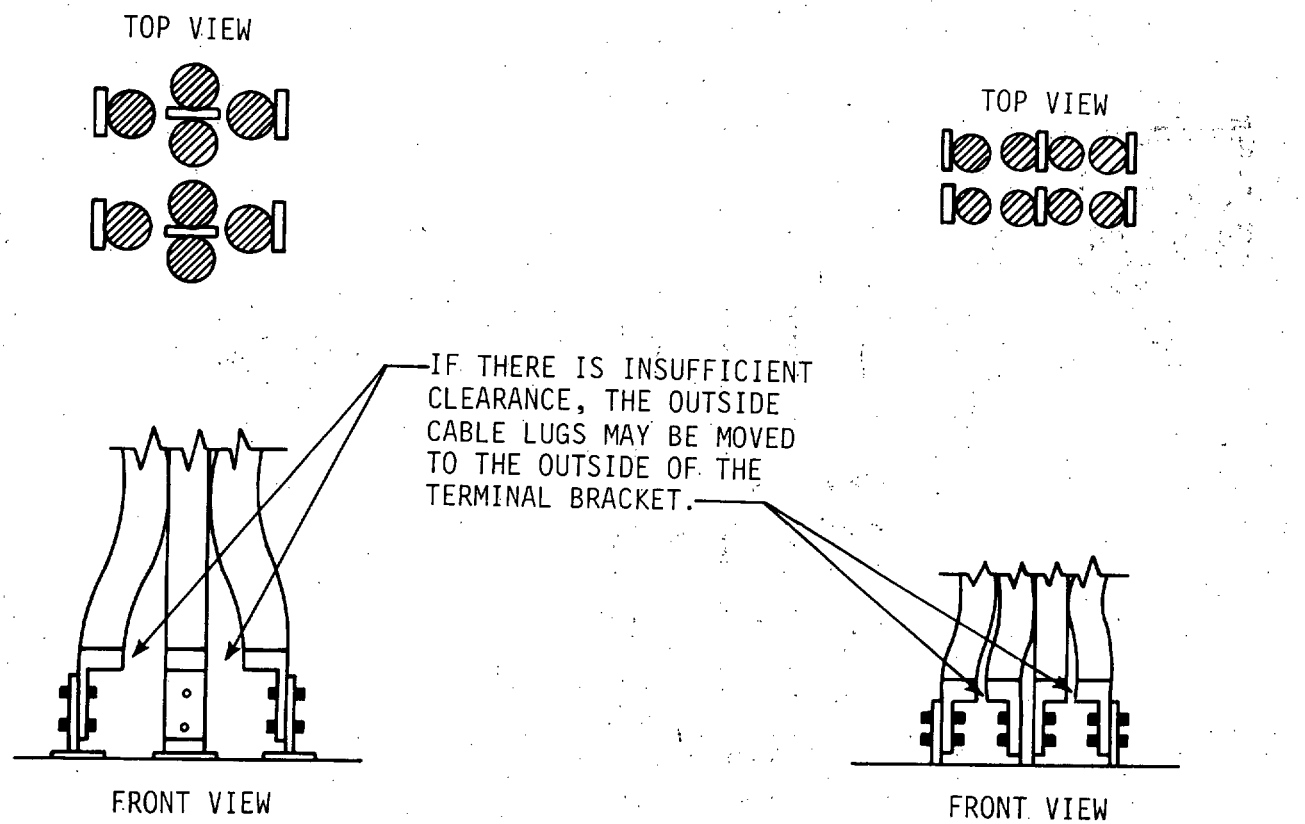
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SH 11957

FIGURE 17. Four-cable, two-terminal lugs in parallel.

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A. TWO GROUPS OF THREE TERMINAL BLADES (SIX TERMINAL BLADES TOTAL)

B. THREE SETS OF PARALLEL BLADES (SIX TERMINAL BLADES TOTAL)

SH 11958

FIGURE 18. Eight-cable, six-terminal blade connections.

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